

BidBazaar: Auction Management System

1. Group Members

- **Member 1:** Arush, 2023136, arush23136@iiitd.ac.in
 - **Member 2:** Harshil, 2023235, harshil23235@iiitd.ac.in
 - **Member 3:** Harshit Dawra, 2023237, harshit23237@iiitd.ac.in
 - **Member 4:** Pragya Singh, 2023379, pragya23379@iiitd.ac.in
-

2. Detailed Requirement Description

Overview:

BidBazaar is an online auction management system that facilitates buyers and sellers to participate in auctions where users can buy and sell products through competitive bidding. The system will manage product listings, auctions, bidding processes, and payments. It will also provide features such as auto-bidding, invoice generation, and inventory management for product pickup and delivery.

Requirements:

- **User Management :** A user must register by providing essential details such as their name, email, contact number, and secure password. Users can buy or sell products (i.e a single user can be a seller or a buyer or both)
- **Product Management:** Sellers who wish to list products for auction follow these steps:

Product Configurations: defines categories for the type of products and different features based on different products .

Product details: Maintains information about individual products listed for auction.

Product Inventory: the product inventory tracks pickup, delivery, and payment information and status of the products.

- **Auction management:** BidBazaar supports multiple auction types, including single price auction and an english auction and also tracks if snipping(a bidder who placed a bid at the last minute) happened or not, start time, end time, negotiation made or not.
 - **Bidding Mechanism :** Buyers participate in ongoing auctions by placing bids. Each bid includes a timestamp and the bid amount. The system records all bids and announces the winning bid at the end of the auction, along with a timestamp for the announcement.
 - **Auto Bidding System :** Where buyers can apply auto bid on some products if they want with adjustable step rates defining the maximum bid limit. The system automatically places bids up to the defined maximum amount when other competitors place bids.
 - **Payment & Invoicing :** Generate invoices for both sellers and buyers after successful auctions, and do profit calculations by maintaining payment details, that have a final seller payouts and buyer costs with a receipt for both. If a buyer or seller refuses to take or sell a product after the end of auction the product will belong to the BidBazaar inventory and the seller would be blocked.
-

3. ER Model

[https://www.figma.com/board/3OPgk2ulNapZodOcs0d9P/Entity-relationship-\(ER\)-Diagram-\(Community\)?node-id=0-1&t=Hnmjwts0jRj6Vu1h-1](https://www.figma.com/board/3OPgk2ulNapZodOcs0d9P/Entity-relationship-(ER)-Diagram-(Community)?node-id=0-1&t=Hnmjwts0jRj6Vu1h-1)

Entities & Attributes

1. **Product Configuration:** *Created, Type, Product, Specifications/features (Model, colour, company etc... depends on product type)*
2. **Product Details:** *Product ID, Product Type, Product, Specification value, Latest Auction Status, Winning Bid ID, Buyer ID, Status*
3. **Product Inventory:** *Product ID, pickup address(from seller), Pickup time and date, Amount for the seller, Delivery address, Delivery address/date, Delivery amount, Status, Seller Receipt ID, Buyer Receipt ID,*
4. **Auction** : *Product ID, Auction ID, Auction Starting timestamp, Auction ending timestamp, Auction status, Auction count, Auction type (single, english), auction starting price, Snipping flag, Auction base price, negotiation enabled(true/ false; enabled in the case of single price auction).*
5. **Bid** : *BidID, AuctionID , product ID, user id, Bid timestamp, Bid amount, isWinningBid, WinTimestamp, Step rate(for autobid feature), isAutoBid (for auto bid feature)*
6. **Auto Bid:** *Auto Bid ID, Auction ID, user ID, Starting bid, Max Bid, Step Rate.*
7. **User** : *UserID, Name, Email, Phone, Password, Address, Notification Mode, isBlocked*
8. **Invoice** : *InvoiceID, Invoice Type(Seller or buyer), Receipt ID(Seller or buyer), Product ID, Created Timestamp, Seller Amount(amount to be given to seller), Service Rate(profit), Final seller payout, Buyer payout, Invoice status.*

5. Relationships

1. User - Bid

- A User places multiple Bids (1:m).
- Each Bid is placed by one User.

2. User - Auction

- A User participates in multiple Auctions (1:m).
- Each Auction involves multiple Users (m:n).

3. User - AutoBid

- A User can create multiple AutoBids (1:m).
- Each AutoBid is linked to a single User.

4. Auction - Product

- Each Auction is associated with one Product (1:1).
- A Product can be auctioned multiple times (1:m).

5. Auction - Bid

- Each Auction consists of multiple Bids (1:m).
- Each Bid belongs to a single Auction.

6. Auction - AutoBid

- An Auction can have multiple AutoBids (1:m).
- Each AutoBid is linked to a single Auction.

7. Product - Product Inventory

- Each Product has one corresponding Product Inventory entry (1:1).

8. Invoice - User

- An Invoice is associated with a single User (m:1).
- A User can have multiple Invoices (1:m).

9. Product - Product Details

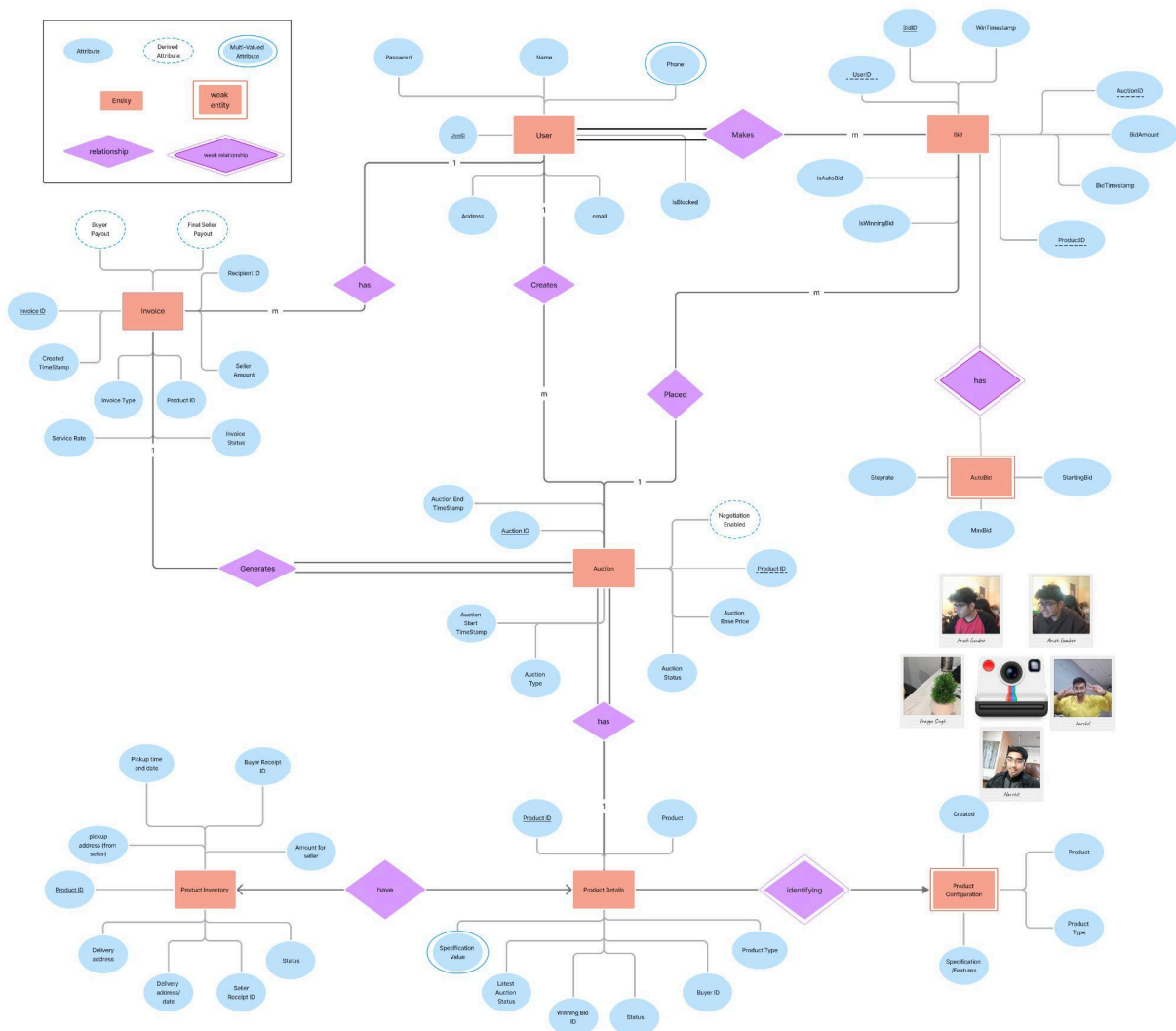
- Each Product has one Product Details entry (1:1).

10. Product Configuration - Product Details

- Each Product Configuration is linked to multiple Product Details (1:m).
-

4. ER Diagram

BidBazar ER Model



5. Comments and Sources

Sources:

- **Class Notes:** for ER Diagrams, and Appropriate Relationships
- **ChatGPT:** for formulating problem statements
- **Online websites** for references (eg. eBay)

Modifications Made: modified relationships management and identify key constraints.
